Practice Problems 1: Lectures 1–6

BUS 411 — Fixed Income Security Analysis and Valuation

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Instructions

These 24 numerical and analytical problems review Lectures 1–6. Four problems are provided for each lecture. Attempt each problem before opening its solution. Keep rates in decimal form during calculations and state monetary units.

1 Lecture 1: Bond Features, Markets, and Risks

1.1 Problem 1: Mapping a Bond's Cash Flows

A corporation issues a \$1,000 par, 6.40% annual-coupon bond with five years to maturity. Coupons are paid semiannually.

a. Compute the semiannual coupon and number of remaining payments.

b. List the cash flows received in periods 1, 9, and 10.

c. Compute total contractual cash flow over the bond's remaining life.

d. Identify the issuer's and investor's obligations.

Solution

The semiannual coupon is

$$C = 1{,}000\left(\frac{0.064}{2}\right) = \boxed{\$32},$$

and there are $5(2) = \boxed{10}$ payments. Periods 1 and 9 each pay \$32; period 10 pays $32+1{,}000 =$ \$1,032. Total contractual cash flow is

$$10(32) + 1{,}000 = \boxed{\$1{,}320}.$$

The issuer must make the promised coupons and repay principal, subject to default. The investor supplies the purchase price and receives the contractual cash flows. Total cash flow is not value because its components arrive at different dates.

1.2 Problem 2: A Partially Amortizing Bond

A \$10 million loan has a 5% annual coupon and repays \$2 million of principal at each year-end for five years. Interest is computed on beginning-of-year principal.

a. Compute interest and total debt service in years 1 and 2.

b. Compute year-5 debt service.

c. Compute total interest over the loan's life.

d. Contrast its principal profile with that of a bullet bond.

Solution

Beginning principal is \$10 million in year 1 and \$8 million in year 2. Therefore,

$$I_1 = 0.05(10) = \$0.50 \text{ million}, \qquad I_2 = 0.05(8) = \$0.40 \text{ million}.$$

Adding the \$2 million amortization gives debt service of \$2.50 million and \$2.40 million. In year 5, beginning principal is \$2 million, so debt service is $0.10 + 2.00 = \boxed{\$2.10 \text{ million}}$. Total interest is

$$0.05(10 + 8 + 6 + 4 + 2) = \boxed{\$1.50 \text{ million}}.$$

A bullet bond repays all principal at maturity. Amortization returns principal earlier, reducing later interest and interest-rate exposure.

1.3 Problem 3: Inflation and Realized Purchasing Power

An investor buys a one-year zero-coupon bond for \$950 and receives \$1,000 at maturity. Inflation over the year is 3.25%.

a. Compute the nominal holding-period return.

b. Compute the exact real return.

c. Compute the maturity payment in beginning-of-year dollars.

d. Name two other risks that this calculation ignores.

Solution

The nominal return is

$$R_n = \frac{1{,}000}{950} - 1 = \boxed{5.2632\%}.$$

The Fisher relation gives

$$R_r = \frac{1 + R_n}{1 + 0.0325} - 1 = \frac{1.052632}{1.0325} - 1 = \boxed{1.9498\%}.$$

The maturity payment has beginning-of-year purchasing power

$$\frac{1{,}000}{1.0325} = \boxed{\$968.52}.$$

The calculation ignores risks such as issuer default, liquidity, reinvestment, exchange-rate, call, and interim price risk.

1.4 Problem 4: Call Risk and Reinvestment

A five-year, \$1,000 par bond pays an 8% annual coupon and is callable at \$1,020 immediately after the year-2 coupon. Just after issuance, comparable yields fall to 4%.

a. Compute the value at year 2 of the remaining three coupon-and-principal cash flows if the bond is not called.

b. Determine whether calling is economically attractive to the issuer.

c. Compute the investor's year-2 cash flow if the bond is called.

d. Explain the investor's reinvestment problem.

Solution

Immediately after the year-2 coupon, the value of the remaining cash flows is

$$P_2 = 80\left(\frac{1-(1.04)^{-3}}{0.04}\right) + 1{,}000(1.04)^{-3} = \boxed{\$1{,}111.00}.$$

Redeeming at \$1,020 rather than leaving debt worth about \$1,111 outstanding is economically attractive, so exercise is likely. At year 2 the investor receives the \$80 coupon plus \$1,020 call price, or $\boxed{\$1{,}100}$. The principal is returned precisely when market yields are low, so replacing the 8% coupon with an equally risky investment is difficult. This is call and reinvestment risk.

2 Lecture 2: Bond Pricing and Yield Measures

2.1 Problem 5: Present Value and an Annuity

An investor will receive \$4,000 at each year-end for four years and an additional \$20,000 with the fourth payment. The annual discount rate is 6%.

a. Compute the present value of the four-payment annuity.

b. Compute the present value of the terminal amount.

c. Compute total present value.

d. Explain why discounting the total \$36,000 for four years is incorrect.

Solution

The annuity value is

$$PV_A = 4{,}000\left(\frac{1-(1.06)^{-4}}{0.06}\right) = \boxed{\$13{,}860.42}.$$

The terminal amount is worth

$$PV_F = 20{,}000(1.06)^{-4} = \boxed{\$15{,}841.87}.$$

Thus total present value is $\boxed{\$29{,}702.29}$. Discounting \$36,000 for four years would incorrectly place the first three \$4,000 payments at year 4.

2.2 Problem 6: Pricing a Semiannual Coupon Bond

A \$1,000 par bond has a 5% annual coupon, six years to maturity, and a 6.20% yield to maturity quoted with semiannual compounding.

a. Identify the periodic coupon, periodic yield, and number of periods.

b. Compute the bond price.

c. State whether it trades at a premium or discount and why.

d. Compute its current yield.

Solution

The inputs are $C = \$25$, $i = 0.062/2 = 0.031$, and $n = 12$. Hence,

$$P = 25\left(\frac{1-(1.031)^{-12}}{0.031}\right) + 1{,}000(1.031)^{-12} = \boxed{\$940.63}.$$

It trades at a discount because its 5% coupon rate is below the required 6.20% yield. Its current yield is

$$\frac{50}{940.63} = \boxed{5.3156\%}.$$

Current yield ignores capital gain and the timing and reinvestment of coupons.

2.3 Problem 7: Clean Price, Accrued Interest, and Invoice Price

A \$1,000 par bond with a 7.20% annual coupon pays semiannually. Settlement occurs 73 days into a 182-day coupon period. The quoted clean price is 101.375% of par.

a. Compute the semiannual coupon.

b. Compute accrued interest using the actual-period fraction.

c. Compute the clean price in dollars.

d. Compute the full (invoice) price.

Solution

The coupon is $1{,}000(0.072/2) = \$36$. Accrued interest is

$$AI = 36\left(\frac{73}{182}\right) = \boxed{\$14.44}.$$

The clean price is $1.01375(1{,}000) = \boxed{\$1{,}013.75}$. Therefore,

$$P_{full} = 1{,}013.75 + 14.44 = \boxed{\$1{,}028.19}.$$

The quoted price excludes accrued interest; the buyer pays the full price at settlement.

2.4 Problem 8: Horizon Return and Reinvestment

A two-year, \$1,000 par bond pays a 6% annual coupon and is bought for \$985. The first coupon is reinvested for one year at 3%; the bond is held to maturity.

a. Compute the horizon value of the first coupon.

b. Compute total wealth at the two-year horizon.

c. Compute the two-year holding-period return.

d. Compute the annualized compound return.

Solution

The first coupon grows to

$$60(1.03) = \boxed{\$61.80}.$$

At maturity the second coupon and principal provide \$1,060, so horizon wealth is $1{,}121.80$. The two-year holding-period return is

$$\frac{1{,}121.80}{985} - 1 = \boxed{13.8883\%}.$$

The annualized return is

$$\left(\frac{1{,}121.80}{985}\right)^{1/2} - 1 = \boxed{6.7185\%}.$$

The result depends on the realized coupon reinvestment rate; YTM embeds a specific reinvestment assumption.

3 Lecture 3: Duration and Convexity

3.1 Problem 9: Price Value of a Basis Point

A bond is worth 103.18 when its yield is 4.99% and 103.02 when its yield is 5.01%. Its face amount is \$5 million.

a. Estimate PVBP per \$100 face value using a central difference.

b. Estimate modified duration using the midpoint price of 103.10.

c. Compute position DV01.

d. Estimate the loss for a 12 bp yield rise using duration only.

Solution

The 2 bp price difference is \$0.16, so the one-basis-point price value is

$$PVBP = \frac{103.18 - 103.02}{2} = \boxed{0.08 \text{ per } \$100}.$$

Since $PVBP \approx D_{mod} P(0.0001)$,

$$D_{mod} = \frac{0.08}{103.10(0.0001)} = \boxed{7.7595}.$$

There are 5,000,000/100 = 50,000 units, so position DV01 is $0.08(50{,}000) = \boxed{\$4{,}000}$. A 12 bp rise produces an estimated loss of $12(\$4{,}000) = \boxed{\$48{,}000}$.

3.2 Problem 10: Duration and Convexity Approximation

A bond has price 96.50, modified duration 6.40, and convexity 52. Its yield rises by 75 bps.

a. Compute the duration-only percentage price change.

b. Compute the convexity adjustment.

c. Compute the combined percentage and dollar price changes.

d. Explain the sign of the convexity term.

Solution

With $\Delta y = 0.0075$,

$$\left.\frac{\Delta P}{P}\right|_D = -6.40(0.0075) = \boxed{-4.8000\%}.$$

The convexity adjustment is

$$\frac{1}{2}(52)(0.0075)^2 = \boxed{0.14625\%}.$$

Thus, $\Delta P/P \approx -4.65375\%$ and

$$\Delta P = 96.50(-0.0465375) = \boxed{-4.4909},$$

giving an estimated price of 92.0091. For an option-free bond with positive convexity, curvature raises the price relative to the tangent-line estimate for either direction of the yield change.

3.3 Problem 11: Portfolio Duration and a Hedge

A \$20 million portfolio holds 30% in a duration-2.0 bond, 45% in a duration-6.0 bond, and 25% in a duration-10.0 bond. A Treasury futures contract has DV01 \$82.

a. Compute portfolio modified duration.

b. Compute portfolio DV01.

c. Compute the number of contracts for a duration hedge.

d. State the appropriate futures position for hedging a long bond portfolio.

Solution

Market-value-weighted duration is

$$D_P = 0.30(2.0) + 0.45(6.0) + 0.25(10.0) = \boxed{5.80}.$$

Therefore,

$$DV01_P = 20{,}000{,}000(5.80)(0.0001) = \boxed{\$11{,}600}.$$

The hedge requires

$$N = \frac{11{,}600}{82} = 141.46,$$

or about $\boxed{141 \text{ or } 142}$ contracts depending on the desired rounding. The portfolio should short futures because a rate rise lowers both the bonds and, typically, the futures price.

3.4 Problem 12: Key-Rate Duration Exposure

A \$12 million portfolio has 2-year, 5-year, and 10-year key-rate durations of 0.80, 3.10, and 2.40. The corresponding key rates change by +15, -5, and +20 bps.

a. Compute the first-order percentage price contribution from each key rate.

b. Compute the total estimated percentage price change.

c. Compute the dollar price change.

d. Explain why total duration alone cannot reproduce every such scenario.

Solution

Using $\Delta P/P \approx -\sum_i KRD_i \Delta y_i$, the contributions are

$$-0.80(0.0015) = -0.1200\%,$$

$$-3.10(-0.0005) = +0.1550\%,$$

$$-2.40(0.0020) = -0.4800\%.$$

The total is $\boxed{-0.4450\%}$, giving

$$\Delta P = 12{,}000{,}000(-0.00445) = \boxed{-\$53{,}400}.$$

Total duration collapses curve exposure into one parallel-shift measure. Key-rate durations retain information about where the curve moves.

4 Lecture 4: Yield Curves, Spot Rates, and Spreads

4.1 Problem 13: Yield Decomposition and Spread Change

A five-year corporate bond yields 6.35% when the matched Treasury yields 4.60%. One month later, the corporate yield is 6.10% and the Treasury yield is 4.20%.

a. Compute the initial benchmark spread.

b. Compute the new benchmark spread.

c. Decompose the corporate yield change into base-rate and spread changes.

d. Interpret the spread movement.

Solution

Initially,

$$s_0 = 6.35\% - 4.60\% = \boxed{175 \text{ bps}}.$$

The new spread is $6.10\% - 4.20\% = \boxed{190 \text{ bps}}$. The Treasury base rate fell 40 bps while the spread widened 15 bps, producing a net corporate yield change of $-40 + 15 = \boxed{-25 \text{ bps}}$. The wider spread indicates weaker relative valuation, liquidity, or perceived credit quality even though the bond's all-in yield declined.

4.2 Problem 14: Bootstrapping Spot Rates

One-year and two-year annual-pay par Treasury securities have coupon rates of 4.00% and 5.00%, respectively. Both have par value 100.

a. Compute the one-year spot rate.

b. Write the two-year par-bond pricing equation.

c. Bootstrap the two-year spot rate.

d. Price a two-year zero-coupon Treasury with face value \$1,000.

Solution

The one-year par security implies $z_1 = \boxed{4.00\%}$. For the two-year bond,

$$100 = \frac{5}{1.04} + \frac{105}{(1+z_2)^2}.$$

Thus,

$$(1+z_2)^2 = \frac{105}{100 - 5/1.04} = 1.10303,$$

so $z_2 = \boxed{5.0253\%}$. The zero's price is

$$P_0 = \frac{1{,}000}{(1.050253)^2} = \boxed{\$906.59}.$$

Each cash flow is discounted at the spot rate matching its date.

4.3 Problem 15: An Implied Forward Rate

The one-year spot rate is 3.20% and the two-year spot rate is 4.10%, with annual compounding.

a. Compute the one-year forward rate one year from now.

b. Verify the result by comparing two-year growth factors.

c. Compute the forward rate if the two-year spot rate instead equals 3.00%.

d. Explain what an implied forward rate is and is not.

Solution

No-arbitrage requires

$$(1+z_2)^2 = (1+z_1)(1+f_{1,1}).$$

Therefore,

$$f_{1,1} = \frac{(1.041)^2}{1.032} - 1 = \boxed{5.0078\%}.$$

Indeed, $(1.032)(1.050078) = (1.041)^2$. If $z_2 = 3.00\%$,

$$f_{1,1} = \frac{(1.03)^2}{1.032} - 1 = \boxed{2.8004\%}.$$

The forward rate is the future borrowing/lending rate locked in by current spot prices. It is not necessarily an unbiased forecast of the future spot rate because risk premiums and other effects may matter.

4.4 Problem 16: A Duration-Neutral Curve Trade

A manager expects the 10-year yield to fall relative to the 2-year yield. A \$4 million long 10-year position has modified duration 8.20. The 2-year bond used for the short hedge has modified duration 1.85.

a. Compute the long position's DV01.

b. Find the market value of the 2-year short position that makes parallel DV01 approximately neutral.

c. If the 10-year yield falls 12 bps and the 2-year yield falls 4 bps, estimate the trade's P&L.

d. Identify one residual risk.

Solution

The long position's DV01 is

$$4{,}000{,}000(8.20)(0.0001) = \boxed{\$3{,}280}.$$

Set the short bond's DV01 equal to \$3,280:

$$MV_S = \frac{3{,}280}{1.85(0.0001)} = \boxed{\$17.7297 \text{ million}}.$$

The long gains $12(3,280) = \$39,360$. The short loses $4(3,280) = \$13,120$, so net P&L is $\boxed{\$26,240}$. Residual risks include convexity, imperfect key-rate mapping, financing, liquidity, and spread changes.

5 Lecture 5: Treasury and Corporate Debt Instruments

5.1 Problem 17: Treasury Bill Pricing

A 182-day Treasury bill has face value \$1,000,000 and a bank discount yield of 4.80%. Use a 360-day year for the discount quotation.

a. Compute the dollar discount.

b. Compute the purchase price.

c. Compute the holding-period yield.

d. Compute the bond-equivalent yield using a 365-day year.

Solution

The dollar discount is

$$D = 1{,}000{,}000(0.048)\left(\frac{182}{360}\right) = \boxed{\$24{,}266.67}.$$

Price is $\boxed{\$975{,}733.33}$. The holding-period yield is

$$HPY = \frac{24{,}266.67}{975{,}733.33} = \boxed{2.4870\%}.$$

The simple bond-equivalent yield is

$$BEY = HPY\left(\frac{365}{182}\right) = \boxed{4.9877\%}.$$

The bank discount yield divides by face value and uses a 360-day year, so it is not directly comparable with a price-based investment yield.

5.2 Problem 18: Treasury Auction Allocation

At a Treasury auction, \$30 billion is offered. Noncompetitive tenders total \$2 billion. Competitive bids at or below the stop-out yield total \$35 billion.

a. How much remains for competitive bidders?

b. Compute the competitive allocation percentage.

c. Find the allocation to a successful \$500 million competitive bid.

d. State the pricing implication of a single-price auction.

Solution

After noncompetitive awards, competitive bidders can receive

$$30 - 2 = \boxed{\$28 \text{ billion}}.$$

The pro rata allocation percentage is

$$\frac{28}{35} = \boxed{80\%}.$$

A successful \$500 million bid receives $500(0.80) = \boxed{\$400 \text{ million}}$. In a single-price auction, accepted competitive and noncompetitive awards receive the price corresponding to the stop-out yield, rather than each bidder's submitted yield.

5.3 Problem 19: Expected Credit Loss and Recovery

A one-year corporate note has \$8 million face value, a 2.5% probability of default, and a 40% recovery rate on face value if default occurs. Ignore coupons and discounting.

a. Compute loss given default in dollars.

b. Compute expected credit loss.

c. Compute the expected maturity principal payment.

d. Recompute expected loss if recovery falls to 25%.

Solution

Loss given default is

$$LGD = 8{,}000{,}000(1 - 0.40) = \boxed{\$4{,}800{,}000}.$$

Expected loss is

$$EL = 0.025(4{,}800{,}000) = \boxed{\$120{,}000}.$$

Expected principal is $8{,}000{,}000 - 120{,}000 = \boxed{\$7{,}880{,}000}$. At 25% recovery, expected loss becomes

$$0.025[8{,}000{,}000(0.75)] = \boxed{\$150{,}000}.$$

Expected loss combines default probability, exposure, and loss severity; it does not describe unexpected loss or mark-to-market spread volatility.

5.4 Problem 20: Commercial Paper Funding Cost

A company issues \$25 million face value of 90-day commercial paper at a price of 99.10% of face value. Ignore issuance fees.

a. Compute cash proceeds.

b. Compute dollar interest cost.

c. Compute the 90-day holding-period yield to the investor.

d. Annualize that yield on a simple 360-day basis and identify the main credit feature of commercial paper.

Solution

The issuer receives

$$25{,}000{,}000(0.991) = \boxed{\$24{,}775{,}000}.$$

Dollar interest is $25{,}000{,}000 - 24{,}775{,}000 = \boxed{\$225{,}000}$. The investor's holding-period yield is

$$HPY = \frac{225{,}000}{24{,}775{,}000} = \boxed{0.9082\%}.$$

The simple annualized yield is $0.9082\%(360/90) = \boxed{3.6327\%}$. Commercial paper is an unsecured short-term corporate obligation, so repayment depends on the issuer's credit and liquidity access.

6 Lecture 6: Bond Portfolio Construction

6.1 Problem 21: Two-Asset Portfolio Risk

A portfolio invests 60% in Bond A and 40% in Bond B. Annual return volatilities are 5% and 8%, and their correlation is 0.25. Expected returns are 4% and 6%.

a. Compute expected portfolio return.

b. Compute portfolio variance.

c. Compute portfolio volatility.

d. Compare it with the weighted average volatility and explain the difference.

Solution

Expected return is

$$E[R_P] = 0.60(0.04) + 0.40(0.06) = \boxed{4.80\%}.$$

Variance is

$$\sigma_P^2 = (0.60)^2(0.05)^2 + (0.40)^2(0.08)^2 + 2(0.60)(0.40)(0.25)(0.05)(0.08) = \boxed{0.002404}.$$

Thus $\sigma_P = \sqrt{0.002404} = \boxed{4.9031\%}$. Weighted average volatility is $0.60(5\%) + 0.40(8\%) = 6.20\%$. Portfolio risk is lower because imperfectly correlated returns provide diversification.

6.2 Problem 22: Ex Post Tracking Error

A portfolio's six monthly active returns, in basis points, are

$$12,\ -8,\ 5,\ -3,\ 10,\ -4.$$

a. Compute mean monthly active return.

b. Compute sample monthly tracking error.

c. Annualize tracking error using $\sqrt{12}$.

d. Explain what tracking error measures.

Solution

The mean is

$$\bar{r}_A = \frac{12 - 8 + 5 - 3 + 10 - 4}{6} = \boxed{2 \text{ bps}}.$$

Deviations are $10, -10, 3, -5, 8, -6$, whose squared sum is 334. Therefore,

$$TE_m = \sqrt{\frac{334}{6-1}} = \boxed{8.1731 \text{ bps per month}}.$$

Annualized tracking error is

$$TE_a = 8.1731\sqrt{12} = \boxed{28.31 \text{ bps per year}}.$$

Tracking error measures the variability of returns relative to a benchmark, not the portfolio's total return volatility or average active return.

6.3 Problem 23: Cell-Based Sampling

A benchmark and a sampled portfolio have the following weights:

Cell	Benchmark	Portfolio	Cell duration
Treasury 1–3 year	25%	20%	1.9
Treasury 7–10 year	30%	35%	7.8
Corporate 1–3 year	20%	25%	2.1
Corporate 7–10 year	25%	20%	7.2

a. Compute each active cell weight.

b. Verify that active weights sum to zero.

c. Compute benchmark and portfolio duration.

d. Identify the portfolio's principal sector and curve tilts.

Solution

Portfolio minus benchmark active weights are

$$\boxed{-5\%, \quad +5\%, \quad +5\%, \quad -5\%},$$

which sum to zero. Benchmark duration is

$$D_B = 0.25(1.9) + 0.30(7.8) + 0.20(2.1) + 0.25(7.2) = \boxed{5.035}.$$

Portfolio duration is

$$D_P = 0.20(1.9) + 0.35(7.8) + 0.25(2.1) + 0.20(7.2) = \boxed{5.075}.$$

The portfolio overweights long Treasuries and short corporates while underweighting short Treasuries and long corporates. Its net duration is only 0.040 above the benchmark, showing that similar aggregate duration can conceal meaningful cell tilts.

6.4 Problem 24: Factor Tracking Error and Risk Contribution

A portfolio has active exposure 0.20 to a level factor and -0.15 to a spread factor. Annual factor volatilities are 4% and 6%, their correlation is 0.30, and annual idiosyncratic tracking-error volatility is 0.50%.

a. Compute the level-factor variance contribution.

b. Compute the spread-factor variance contribution and covariance term.

c. Compute total tracking-error variance and volatility.

d. Explain one way the manager could reduce predicted tracking error.

Solution

Let active exposures be $b_L = 0.20$ and $b_S = -0.15$. The diagonal contributions are

$$b_L^2 \sigma_L^2 = (0.20)^2 (0.04)^2 = 0.000064,$$

$$b_S^2 \sigma_S^2 = (-0.15)^2 (0.06)^2 = 0.000081.$$

The covariance term is

$$2 b_L b_S \rho \sigma_L \sigma_S = 2(0.20)(-0.15)(0.30)(0.04)(0.06) = -0.0000432.$$

Adding idiosyncratic variance, $(0.005)^2 = 0.000025$, gives

$$TE^2 = 0.000064 + 0.000081 - 0.0000432 + 0.000025 = \boxed{0.0001268},$$

so $TE = \sqrt{0.0001268} = \boxed{1.1261\%}$. The manager can move factor exposures toward zero, substitute securities with better benchmark matching, or reduce idiosyncratic concentration, subject to return and transaction-cost objectives.